

# L1.9 L'Hôpital's Rule

Work in your group. Probe every limit before you touch it.

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## Where L'Hôpital's Rule comes from

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1. Two functions are both zero at  $x = 3$ . Their slopes there are  $f'(3) = 10$  and  $g'(3) = -4$ .

(a) Step off  $x = 3$  by a nudge  $dx$ . Write  $df$  and  $dg$ .

(b) What is  $\lim_{x \rightarrow 3} \frac{f(x)}{g(x)}$ ?

(c) Your answer to (b) has no  $dx$  in it. Why does that matter?

2. Evaluate each of these.

(a)  $\lim_{x \rightarrow 0} \frac{5x}{x}$

(b)  $\lim_{x \rightarrow 0} \frac{x^2}{x}$

(c)  $\lim_{x \rightarrow 0} \frac{x}{x^3}$

The two direct forms:  $\frac{0}{0}$  and  $\frac{\infty}{\infty}$

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3.  $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$

4.  $\lim_{x \rightarrow \infty} \frac{2x^2 - 5x + 1}{6x^2 + x + 3}$

5.  $\lim_{x \rightarrow -2} \frac{x + 2}{x^2 + 3x + 2}$  — do it **two ways**.

6.  $\lim_{x \rightarrow 0} \frac{x^2}{1 - \cos x}$

**Stop.** Check the form again after every application. When it stops being indeterminate, stop.

7.  $\lim_{x \rightarrow \infty} \frac{\sqrt{4x^2 + 1}}{3x - 1}$

8. For which of these is L'Hôpital's Rule **allowed**? Probe each one first, and write the form you got.

(a)  $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$

(b)  $\lim_{x \rightarrow 0} \frac{\cos x}{x^2}$

(c)  $\lim_{x \rightarrow \pi} \frac{\sin x}{x - \pi}$

(d)  $\lim_{x \rightarrow 1} \frac{x^2 + 3}{x + 1}$

### At the boards

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9. **BOARDS** Build a limit to order.

(a) Write a limit as  $x \rightarrow 0$  that the diagnostic would find a signature of  $\frac{0}{0}$  for, but whose value is **7**.

(b) Now one whose diagnostic is  $\frac{0}{0}$  and whose value is **0**.

(c) Now one whose diagnostic is to  $\frac{0}{0}$  and needs the rule **twice**.